Customer Insights and Data Analytics Early Remote Internship

Week-3 Deliverable

CUSTOMER SUPPORT DASHBOARD INSIGHTS, ACTIONABLE RECOMMENDATIONS & TICKET FORECASTING REPORT

Team-2

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1. Introduction

The Week 3 **"Customer Support Dashboard Insights, Actionable Recommendations & Ticket Forecasting Report"** activity focuses on analyzing the cleaned Freshdesk customer support dataset from Week 2 to uncover meaningful trends and insights. This phase emphasizes understanding operational patterns, identifying recurring issues, and translating raw ticket data into actionable knowledge that can improve both service efficiency and customer experience. Trend analysis in customer support is critical because it allows teams to anticipate workload challenges, detect bottlenecks, and optimize processes before they impact customer satisfaction. By examining ticket volumes, response times, agent performance, and sentiment patterns, we can identify areas for improvement and recommend strategies to enhance overall support operations.

Our aim is:

- To create a **visual dashboard** summarizing key ticket trends, categories, priorities, response/resolution times, agent workload, and sentiment.
- To identify trends in **ticket volume, issue types, and priorities** to understand customer challenges.
- To analyze **response and resolution efficiency** and detect delays across categories and agents.
- To evaluate **agent workload distribution** and its impact on ticket handling and customer experience.
- To assess **customer sentiment** and link it to issue types, delays, and support quality.
- To generate **actionable recommendations** to improve workflows, reduce recurring issues, and optimize service delivery.
- To **forecast ticket volumes** based on observed trends for proactive resource allocation and planning.

2. Structured Dataset Review

This section outlines how we prepared and organized the customer support ticket data for analysis. It explains the progression from the originally assigned dataset to the filled version and finally to a cleaned and structured dataset suitable for identifying trends and insights. The section also clarifies the structure, scope, and key characteristics of the dataset to ensure transparency and reliability in the analysis that follows.

2.1 Dataset Reference Overview

Dataset Type	**Dataset Name (links)**
Assigned Dataset	Freshdesk Customer Support Tickets Dataset
Filled Dataset (Week-1 Output)	Freshdesk Customer Support Tickets Dataset Filled
Cleaned & Structured Dataset (Week-2 Output)	Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled.csv

2.2 Structured Dataset Description

The table below summarizes each column, its datatype, and the method used to clean or structure it:

Column Name	**Datatype & Explanation**	**Cleansing Method/Description**
Ticket ID	Object (text/string) – each ticket has a unique ID stored as text	Already unique identifiers, no cleaning needed
Created Date	Datetime – date when ticket was submitted	Converted to standard datetime datatype format, checked for missing

		values
Resolved Date	Datetime – date when ticket was resolved	Converted to standard datetime datatype format, checked for missing values
Category	Object (text/string) – ticket type like Technical, Billing	Checked missing & standardization of categories but were already structured
Priority	Object (text/string) – ticket urgency level	Checked missing, standardization/typos of priority levels (High, Medium, Low), but were correct already
Status	Object (text/string) – ticket stage (Open/Resolved/Closed)	Ensured no missing & consistency of status labels (Resolved, Closed)
Agent	Object (text/string) – name of the assigned support agent	Checked missing of agent names; were already in structured
Tags	Object (text/string) – labels describing ticket details	Checked Parsed multiple tags into structured format & irrelevant/empty tags, but already were fine.
Sentiment	Object (text/string) – Positive, Neutral, Negative customer sentiment	Positive, Negative sentiments were fine . Only 'Neutral ' with extra spaced was found in several rows which was corrected & replaced with no space of 'Neutral' sentiment.
Subject/ Short Description	Object (text/string) – brief text describing the ticket	Was already filled in every row, didn't needed for any detection of cleaning.

Additional Notes:

- The cleaned dataset contains 30 customer support tickets, each assigned a unique Ticket ID ranging from T011 to T040, ensuring that no two tickets share the same identifier. This range indicates a continuous and sequential logging of tickets during the monitored period.
- The time frame covered by the dataset is from January 4 to January 10, 2026, capturing one full week of ticket activity. This allows for the identification of daily trends in ticket

submissions, peak activity periods, and resolution patterns over consecutive days. By analyzing this time span, it is possible to assess workload distribution and responsiveness within the support team.

- The dataset has a dimension of 30 × 11, meaning it contains 30 rows (tickets) and 11 columns (variables). Each column represents a specific attribute of the tickets, including operational, time-based, and experience-related data points. The dataset is structured to facilitate both quantitative analysis, such as response time calculation, and qualitative analysis, such as sentiment and tag-based insights.

2.3 Key Variables Used for Analysis

1. **Time-based Variables**

- *Created Date:* Captures when each ticket was logged; used to identify trends over time, such as peak ticket submission periods.
- *Resolved Date:* Captures when tickets were resolved; enables calculation of response/resolution times and backlog patterns.

2. **Operational Variables**

- *Category:* Classifies tickets by issue type; helps in identifying recurring problems and areas requiring operational improvement.
- *Priority:* Indicates urgency (High, Medium, Low); used to assess whether high-priority tickets are addressed promptly.
- *Status:* Tracks progress of tickets (Open, In Progress, Closed); allows monitoring of workflow efficiency.
- *Agent:* Identifies staff handling tickets; useful for workload distribution and performance evaluation.

3. **Experience-based Variables**

- *Tags:* Represents specific ticket attributes or themes; facilitates deeper insights into customer concerns.
- *Sentiment:* Reflects customer satisfaction or dissatisfaction; enables evaluation of service quality and customer experience trends.

Overall, the dataset is complete and well-structured. This makes it suitable for identifying trends, operational bottlenecks, and customer experience patterns efficiently.

3. Dashboard Creation Process & Methodology

3.1 Tool Selection & Design Approach

This section shows how the **Freshdesk Customer Support Ticket Insights & Analytics – Power BI Dashboard** was designed and developed to support trend identification, performance monitoring, and insight generation. We selected **Microsoft Power BI** as the primary tool due to its strong capabilities in interactive visualization, dynamic filtering, and KPI-driven analytics, which are essential for customer support reporting.

We designed the dashboard to present a unified view of support operations by integrating key metrics such as ticket volume, resolution efficiency, SLA compliance, agent workload, and customer sentiment. Interactive KPI cards, gauges, and analytical charts enable both quick performance assessment and deeper trend analysis. We also implemented slicers for **Created Date, Resolved Date, Category, Status, and Sentiment** to allow flexible, user-driven analysis. All visuals and metrics were derived from the **cleaned and structured dataset**, ensuring accuracy and consistency across the dashboard.

3.2 Key Performance Indicators (KPIs) – Metrics & Calculation Logic

Table 1: Key Performance Indicators (KPIs) Details

KPI Title	Chart Type	DAX Formula / Columns Used	Analytical Significance

Customer Satisfaction (CSAT) Score (%)	Gauge	`Positive_Tickets =`
		`CALCULATE(`
		`COUNTROWS('Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'),`
		`'Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'[Sentiment] = "Positive"`
		`)`
		`Resolution Time (Hours) =`
			Represents overall customer satisfaction by measuring the proportion of positive sentiment tickets and benchmarking service quality against a standard target.

		AVERAGEX('Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled', DATEDIFF('Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'[Created Date], 'Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'[Resolved Date], HOUR)) CSAT Target = 80	
Total Tickets	Card	Total Tickets = COUNT ('Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'[Ticket ID])	Indicates the total volume of customer support demand handled during the analysis period.
Resolution Rate (%)	Card	Resolved Tickets = CALCULATE (COUNT ('Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'[Ticket ID]), FILTER (	Evaluates the effectiveness of the support team in resolving incoming issues.

<table>
<tr><td></td><td></td><td><pre>'Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled',

'Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'[Status] = "Resolved"
            ||
'Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'[Status] = "Closed"
)
)
Resolution Rate (%) =
DIVIDE ([Resolved Tickets], [Total Tickets], 0)</pre></td><td></td></tr>
<tr><td>Average Resolution Time (Hours)</td><td>Card</td><td><pre>Avg Resolution Time (Hours) =
AVERAGEX (

'Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled',
    DATEDIFF (

'Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'[Created Date],

'Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'[Resolved Date],
        MINUTE</pre></td><td>Measures operational efficiency by tracking how quickly tickets are resolved on average.</td></tr>
</table>

		) / 60)	
SLA Compliance Rate (%) (≤ 2 Hours)	Gauge	SLA Compliance (%) = DIVIDE (CALCULATE (COUNT ('Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'[Ticket ID]), FILTER ('Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled', DATEDIFF ('Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'[Created Date], 'Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled'[Resolved Date], MINUTE) <= 120)), [Total Tickets], 0)	Assesses adherence to service-level expectations and identifies delays in ticket handling.

3.3 Key Visualizations & Analytical Purpose

Table 2: Key Visualizations Details

Visualization Title	Chart Type	Columns / Fields Used	Analytical Significance
Distribution of Support Tickets by Category	Clustered Column Chart	Axis: Category Values: Count of Ticket ID	Identifies dominant issue types and recurring problem areas.
Ticket Distribution by Priority Level	Donut Chart	Legend: Priority Values: Count of Ticket ID	Highlights the severity and urgency of incoming support requests.
Customer Sentiment Distribution Across Tickets	Clustered Bar Chart	Axis: Sentiment Values: Count of Ticket ID	Provides an overview of customer experience and emotional response.
Ticket Handling Volume by Agent	Pie Chart	Legend: Agent Values: Count of Ticket ID	Evaluates workload distribution among support agents.
Daily Trend of Tickets & Resolution Time	Line Chart (Dual Axis)	X Axis: Created Date (Day) Y Axis: Count of Ticket ID Secondary Y Axis:Resolution Time (Hours)	Tracks fluctuations in ticket volume alongside resolution efficiency over time.
Ticket Volume by Tags	Funnel Chart	Values: Count of Ticket ID Category: Tags	Identifies the most frequent and impactful recurring issue types.

3.4 Filters (Slicers) Configuration

Table 3: Filters (Slicers) Details

Filter Name	Slicer Type	Column Used	Analytical Significance	Expected Analytical Use
Created Date	Date Slicer	Created Date	Enables time-based trend analysis of ticket inflow.	Compares KPIs and ticket volume across selected periods.
Resolved Date	Date Slicer	Resolved Date	Focuses analysis on resolution efficiency over time.	Evaluates SLA compliance and resolution speed.
Status	Dropdown/List	Status	Filters tickets by workflow stage.	Identifies unresolved or delayed tickets.
Sentiment	Dropdown/List	Sentiment	Segments tickets by customer experience.	Analyzes how sentiment affects KPIs and trends.
Category	Dropdown/List	Category	Separates tickets by issue type.	Focuses analysis on specific problem areas.

3.5 Dashboard Coverage Mapping

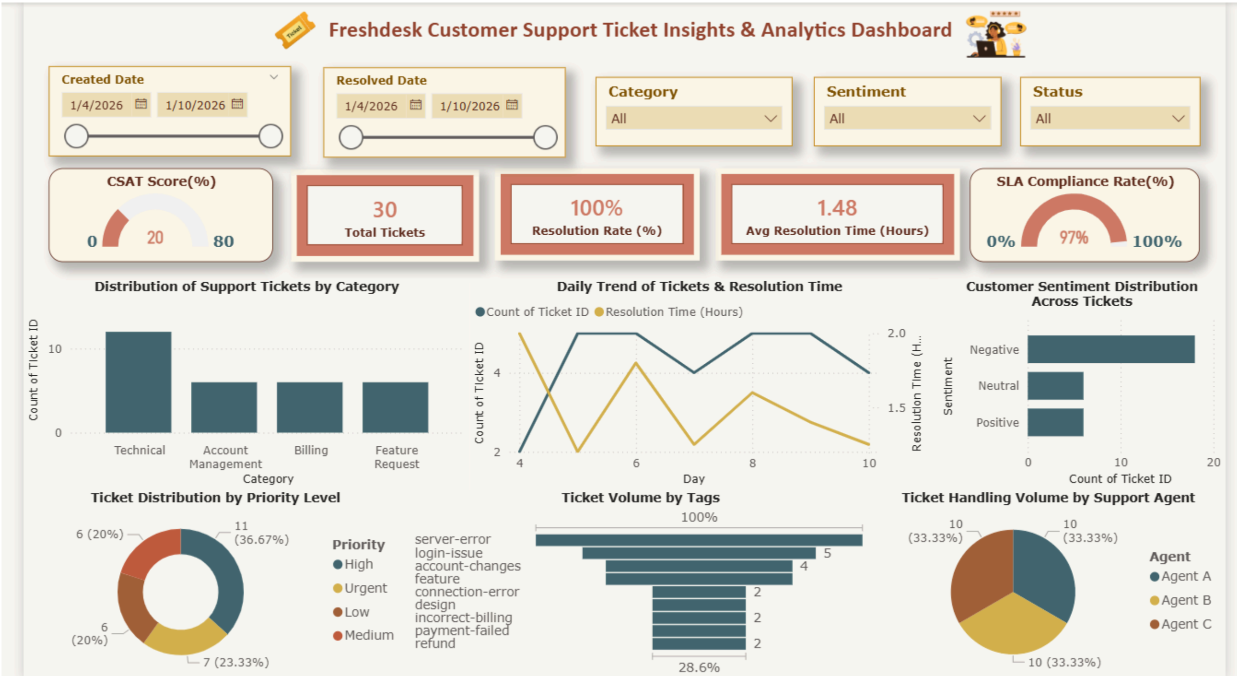
Table 4: Dashboard Coverage Mapping (Filters, KPIs, and Visualizations)

Analysis Focus / Requirement Covered	Filter	KPI	Visualization
Ticket Volume & Time-Based Trends	Created Date, Resolved Date	Average Resolution Time (Hours)	Daily Trend of Tickets & Resolution Time (Line Chart)
Issue Category & Recurring Problems	Category	Total Tickets	Distribution of Support Tickets by Category (Bar Chart)

Priority & Ticket Severity Analysis	Status	Resolution Rate (%)	Ticket Distribution by Priority Level (Donut Chart)
Service Quality & SLA Performance	Resolved Date	SLA Compliance (%)	—
Customer Sentiment & Experience/Satisfaction Impact	Sentiment	Customer Satisfaction (CSAT) Score (%)	Customer Sentiment Distribution Across Tickets (Clustered Bar Chart)
Agent Workload & Performance Efficiency	—	—	Ticket Handling Volume by Agent (Pie Chart)
Recurring Issues & Common Problem Patterns	Category	—	Ticket Volume by Tags (Funnel Chart)

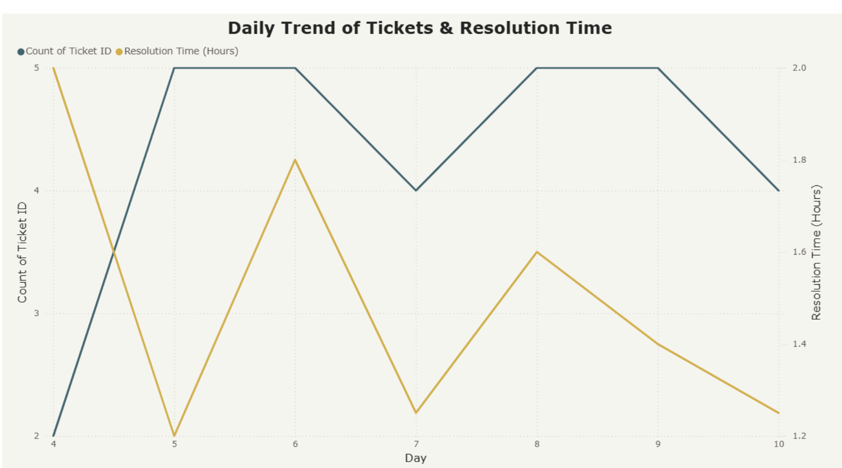
To ensure that the dashboard effectively addresses the analytical objectives of Week 3, we systematically aligned filters, KPIs, and visualizations with key customer support performance areas. Each visualization was deliberately selected to answer specific business questions related to trends, efficiency, workload, and customer experience. This structured design approach ensures comprehensive coverage of insights while maintaining clarity and usability within the dashboard.

4. Dashboard Visual Storytelling & Analytical Findings



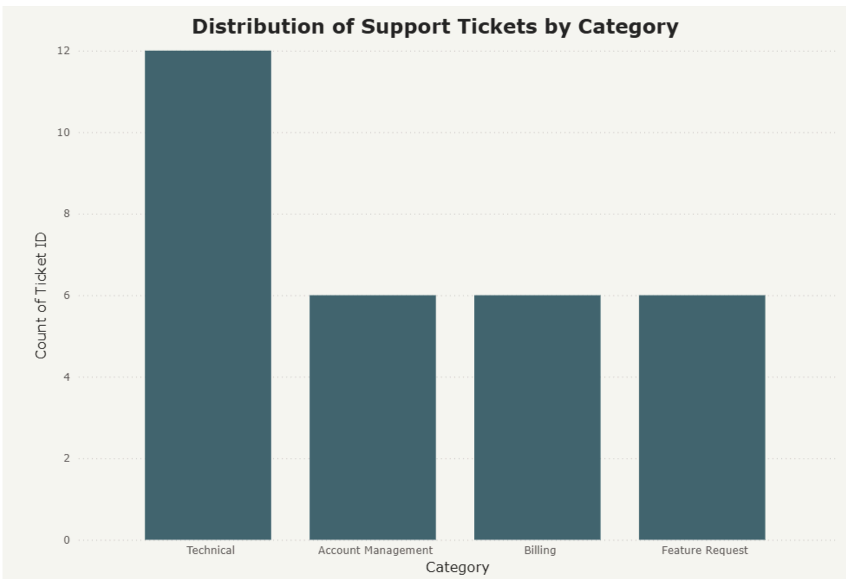
The figure above presents the final ***Freshdesk Customer Support Ticket Insights & Analytics – Power BI Dashboard***, developed to translate structured ticket data into meaningful insights. We designed the dashboard to provide an intuitive, end-to-end view of support operations, enabling quick identification of trends, performance gaps, and customer experience patterns through interactive visuals and KPIs.

4.1 Ticket Volume & Time-Based Trends



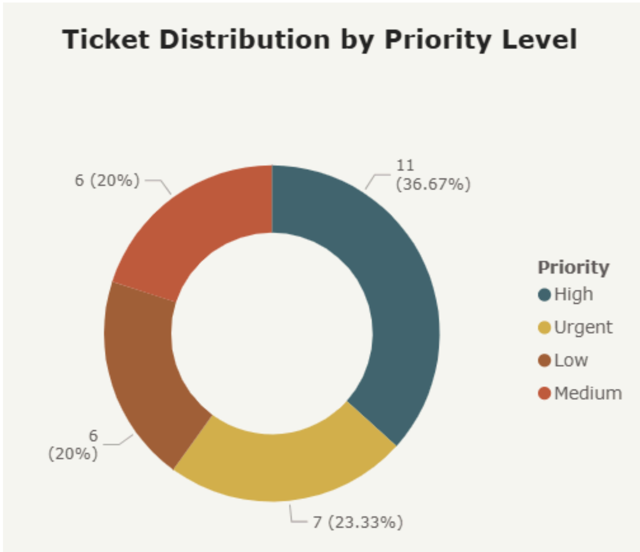
The daily trend chart shows clear fluctuations in ticket volume across the observed period rather than a stable pattern. Ticket counts peak around the middle of the timeline, indicating short-term surges in customer issues. At the same time, average resolution time varies alongside ticket volume, suggesting that higher inflow days place additional pressure on support operations. This relationship implies that increased ticket volume may contribute to slower resolution times. However, resolution time does not spike excessively, indicating that the support system is able to absorb moderate workload increases. This raises the question of whether repeated mid-period spikes are driven by recurring system or product-related issues.

4.2 Issue Category & Recurring Problems



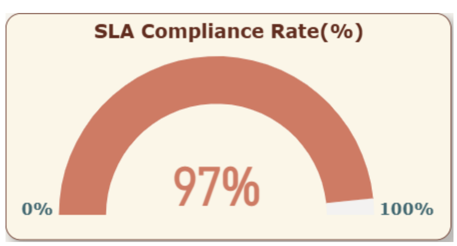
The category distribution chart shows that **Technical issues dominate ticket volume**, followed by Account Management, Billing, and Feature Requests. This concentration indicates that most customer interactions are problem-driven rather than enhancement-driven. The dominance of technical tickets suggests persistent system or access-related challenges faced by users. Billing and account issues also form a consistent portion, reflecting operational or transactional friction. Feature requests remain comparatively lower, indicating fewer proactive or exploratory interactions. This pattern suggests that resolving technical root causes could significantly reduce overall ticket volume.

4.3 Priority & Ticket Severity Analysis



The priority-level donut chart shows that **High and Medium priority tickets form the majority**, while Urgent and Low priorities are less frequent. This distribution indicates that most issues require timely attention but are not immediate emergencies. The presence of urgent tickets confirms the occurrence of critical failures, likely linked to technical or billing problems. Meanwhile, the limited number of low-priority tickets suggests effective prioritization rather than misclassification. This balance reflects a relatively mature ticket triaging process. However, the concentration of higher priorities raises concern about sustained operational load if such issues persist.

4.4 Service Quality & SLA Performance

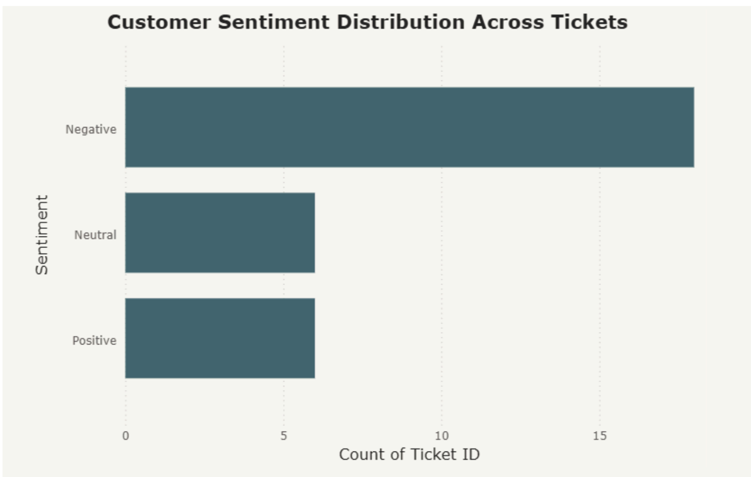


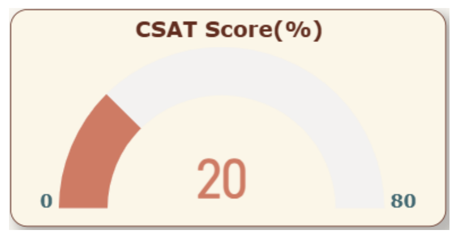
The SLA compliance gauge shows a **high compliance rate of approximately 97%**, indicating strong adherence to service standards. The average resolution time of **1.48 hours** further supports efficient ticket handling. Despite fluctuating ticket volumes, the team maintains consistent service quality. This suggests effective operational processes and agent responsiveness. However, the small proportion of SLA breaches indicates room for targeted improvement, especially during peak volume days. The dashboard raises the question of whether SLA misses are linked to specific categories or high-priority tickets.

➔ **Are delays directly affecting CSAT?**

The high SLA compliance and low average resolution time suggest delays are limited, reducing the likelihood of widespread negative CSAT impact.

4.5 Customer Sentiment & Experience/Satisfaction Impact

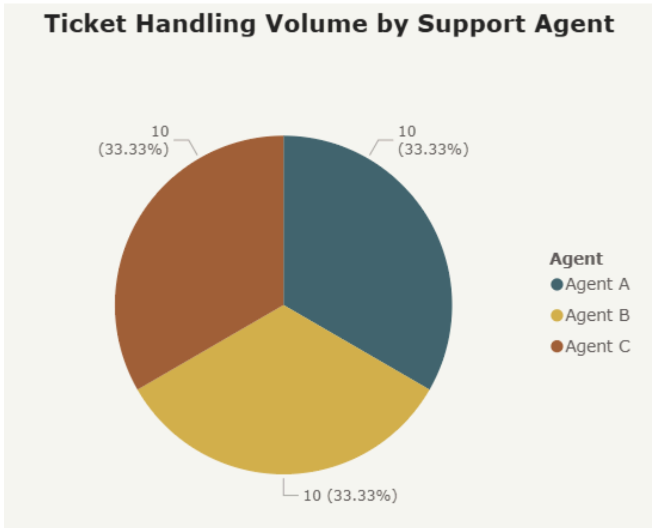




The sentiment distribution chart shows that **negative sentiment is the most prevalent**, followed by neutral and then positive sentiment. Negative sentiment appears closely associated with technical failures and billing issues. Neutral sentiment reflects informational or routine account-related queries. Positive sentiment is largely linked to feature requests and enhancement suggestions. This distribution demonstrates a clear relationship between issue type and customer emotion. The dominance of negative sentiment suggests that unresolved or recurring technical problems significantly affect customer experience.

➔ **Do specific categories consistently receive low ratings?**
Yes, technical and billing-related categories are strongly associated with negative sentiment.

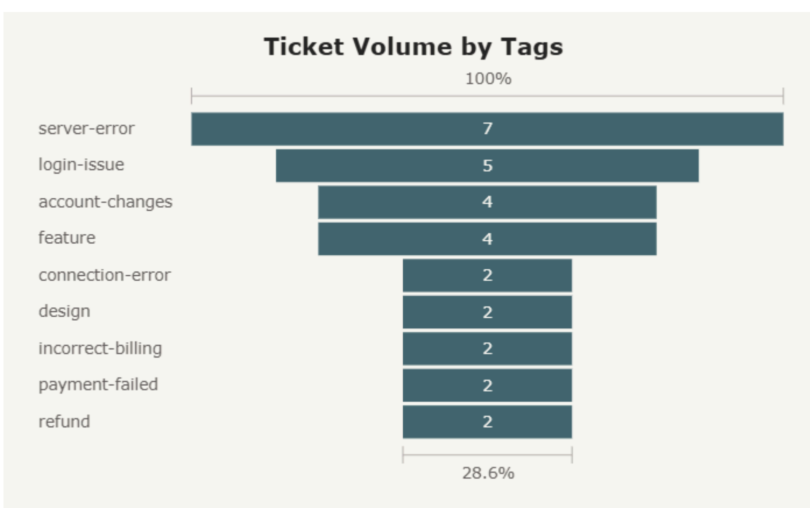
4.6 Agent Workload & Performance Efficiency



The agent workload pie chart shows an **even distribution of tickets across Agent A, B, and C**, each handling roughly one-third of total tickets. This balance indicates effective workload allocation and reduces the risk of agent burnout. Even distribution supports consistent service quality across agents. There is no visible evidence of overworked agents affecting performance. This reflects efficient ticket routing and team management. However, the dashboard does not show ticket complexity, which could further influence workload assessment.

- ➔ **Are overworked agents delivering lower-quality responses?**
  No clear evidence, workload appears evenly distributed among agents.

4.7 Recurring Issues & Common Problem Patterns



The tag frequency chart highlights repeated occurrences of **server-error, login-issues, account-changes, payment-failed, and refund** tags. This repetition confirms that many customer issues are recurring rather than one-time events. Server and login-related problems appear most frequently, indicating systemic reliability concerns. Billing-related tags also recur, suggesting transactional friction. These recurring patterns explain the dominance of technical categories and negative sentiment. The dashboard clearly signals the need for proactive system fixes to prevent repeated complaints.

➔ **Are customers repeatedly reporting the same issues?**
Yes, repeated technical and billing tags confirm recurring customer problems.

5. Actionable Recommendations

5.1 Recommendation 1: Add more focus on the technical and billing teams

- **Observe:** We can observe from sections 4.2, 4.5, and 4.7 show that Technical and Billing category tickets together dominate both negative sentiment tickets and overall tickets, with the top tags being server-error and login issues from the Technical category.

- **Interpret:** Being these tickets are a huge majority of negative sentiments, these categories are the biggest factor lowering CSAT. Issues from both these categories significantly impact operations as well since having a properly running website and on-time payments is absolutely vital to ensuring smooth operations.

- **Recommend:** To help with this, we must dedicate more resources to the Technical and Billing categories. We can do this by implementing specialized agents, proactive system monitoring, and employing a larger team.

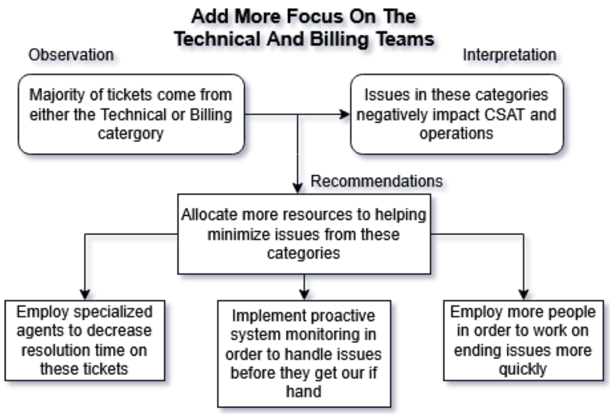
5.2 Recommendation 2: Implementing customer automation

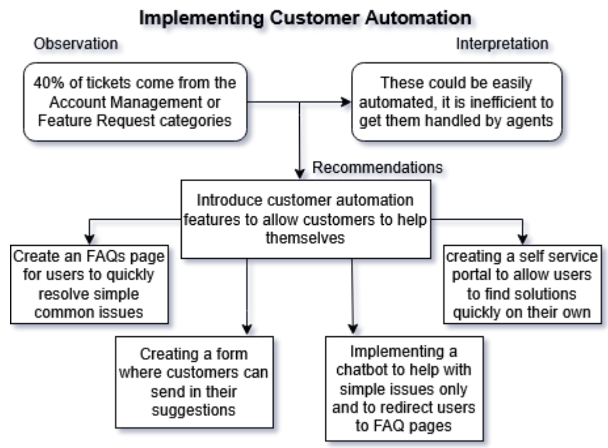
- **Observe:** Section 4.2 shows that the Account Management and Feature Request categories add up to make 40% of total tickets.

- **Interpret:** Issues in these categories are simple, could easily be automated, and do not require agents to handle them. Still getting them handled by agents lowers efficiency and takes away attention from more complex tickets. Lowered efficiency potentially means reducing CSAT.

- **Recommend:** Suggestions to help improve this include featuring FAQs, introducing a suggestions form, adding a chatbot, and implementing a self-service portal.

5.3 Insight Generation & Actionable Recommendation Framework





6. Ticket Volume Forecast

The purpose of this section is to estimate future customer support ticket volumes based on historical data. Forecasting ticket volume helps anticipate support workload, plan staffing and resources, and make data-driven operational decisions.

Two approaches were used for forecasting:

1. **Option A — Weekly Average Forecast**: Assumes a stable ticket volume based on historical weekly averages.

2. **Option B — Linear Trend Forecast**: Projects ticket volume growth using a linear trend observed in historical data.

6.1 Data Preparation

- **Historical tickets** were aggregated on a **weekly basis** using the `Created Date` column.

- The dataset was transformed into a weekly summary table (output):

Step 1: Load dataset and prepare weekly aggregation

```
[1]
✓ 22s
# Imporing necessary libraries
import pandas as pd
import numpy as np

# Google Drive
from google.colab import drive
drive.mount('/content/drive')

# Defining the file path
file_path = '/content/drive/MyDrive/Customer Insights and Data Analytics Early Remote Internship/Cleaned_Freshdesk_Customer_Support_Tickets_Dataset_Filled.csv'

# Reading the CSV file
df = pd.read_csv(file_path)

# Ensuring Created Date is datetime
df['Created Date'] = pd.to_datetime(df['Created Date'])

# Setting Created Date as index (optional)
df.set_index('Created Date', inplace=True)

# Resample by week and count tickets
weekly_tickets = df['Ticket ID'].resample('W').count().reset_index()
weekly_tickets.rename(columns={'Ticket ID': 'Tickets'}, inplace=True)

# Displaying weekly ticket counts
print(weekly_tickets)
```

```
Mounted at /content/drive
  Created Date  Tickets
0   2026-01-04        2
1   2026-01-11       28
```

6.2 Forecast Methods

Option A: Weekly Average Forecast

- Calculated the **average number of tickets per week** from historical data:

Step 2: Option A — Weekly Average Forecast

```
[2]
✓ 0s
# Calculate weekly average
average_tickets = weekly_tickets['Tickets'].mean()
print("Weekly Average Tickets:", round(average_tickets, 2))

# Forecast next 4 weeks using average
forecast_weeks = 4
forecast_avg = pd.DataFrame({
    'Week': range(1, forecast_weeks + 1),
    'Forecasted Tickets (Avg)': [round(average_tickets, 0)] * forecast_weeks
})

print(forecast_avg)
```

```
Weekly Average Tickets: 15.0
   Week  Forecasted Tickets (Avg)
0     1                      15.0
1     2                      15.0
2     3                      15.0
3     4                      15.0
```

Option B: Linear Trend Forecast

- A **linear regression model** was fitted to the historical weekly ticket counts to predict future ticket trends.

- Forecast for the next 4 weeks:

Step 3: Option B — Linear Trend Projection

```
import numpy as np
from sklearn.linear_model import LinearRegression

# Prepare data for linear regression
weekly_tickets['Week_Num'] = np.arange(len(weekly_tickets))  # 0,1,2,...
X = weekly_tickets[['Week_Num']]  # feature
y = weekly_tickets['Tickets']     # target

# Fit linear model
model = LinearRegression()
model.fit(X, y)

# Forecast next 4 weeks
future_weeks = np.arange(len(weekly_tickets), len(weekly_tickets) + forecast_weeks).reshape(-1,1)
forecast_linear = model.predict(future_weeks)

# Create dataframe
forecast_linear_df = pd.DataFrame({
    'Week': range(len(weekly_tickets)+1, len(weekly_tickets)+1+forecast_weeks),
    'Forecasted Tickets (Linear)': np.round(forecast_linear,0)
})

print(forecast_linear_df)
```

```
   Week  Forecasted Tickets (Linear)
0     3                         54.0
1     4                         80.0
2     5                        106.0
3     6                        132.0
```

6.3 Combined Forecast Plotting

- Historical, Average and linear forecast data were combined for visualization purposes:

Step 4: Visualization - Combine Historical + Average + Linear Forecasts

```
import matplotlib.pyplot as plt

# Make a copy for plotting
combined_df_plot = combined_df.copy()

# Ensure 'Resolved Date' is datetime
combined_df_plot['Resolved Date'] = pd.to_datetime(combined_df_plot['Resolved Date'])

# Step 1: Historical Tickets
historical_mask = combined_df_plot['Historical_Tickets'].notna()

# Step 2: Average Forecast
avg_mask = combined_df_plot['Avg_Forecast'].notna()

# Step 3: Linear Forecast
linear_mask = combined_df_plot['Linear_Forecast'].notna()

# Step 4: Plot
plt.figure(figsize=(12,6))
```

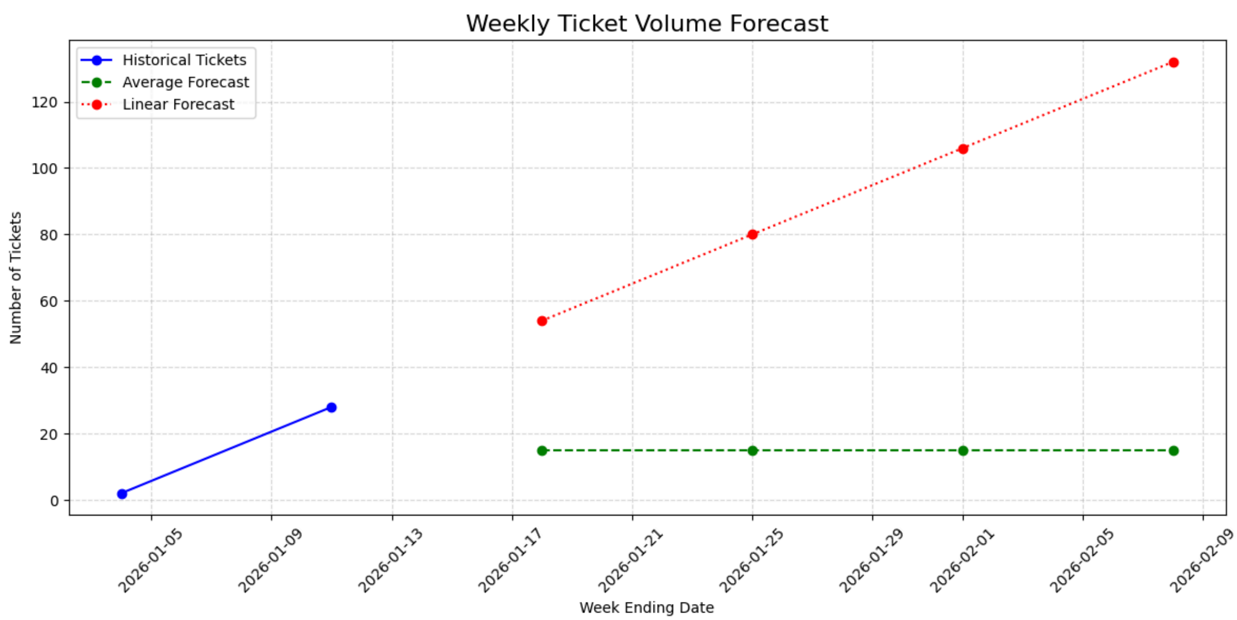
```
# Historical Tickets
plt.plot(combined_df_plot['Resolved Date'][historical_mask],
         combined_df_plot['Historical_Tickets'][historical_mask],
         marker='o', linestyle='-', color='blue', label='Historical Tickets')

# Average Forecast
plt.plot(combined_df_plot['Resolved Date'][avg_mask],
         combined_df_plot['Avg_Forecast'][avg_mask],
         marker='o', linestyle='--', color='green', label='Average Forecast')

# Linear Forecast
plt.plot(combined_df_plot['Resolved Date'][linear_mask],
         combined_df_plot['Linear_Forecast'][linear_mask],
         marker='o', linestyle=':', color='red', label='Linear Forecast')

# Formatting
plt.title('Weekly Ticket Volume Forecast', fontsize=16)
plt.xlabel('Week Ending Date')
plt.ylabel('Number of Tickets')
plt.xticks(rotation=45)
plt.grid(True, linestyle='--', alpha=0.5)
plt.legend()
plt.tight_layout()
plt.show()
```

6.4 Forecast Insight / Visualization

- A **line chart** was created to show historical data alongside both forecast models.

Chart Description:

- **X-axis:** Week ending date (weekly granularity)
- **Y-axis:** Number of tickets
- **Series:**
  - Blue Solid Line → Historical Tickets
  - Green Dashed Line → Average Forecast (stable volume)
  - Red Dotted Line → Linear Forecast (growth-oriented trend)



6.5 Forecast Interpretation & Observations

1. **Historical Trend:** Ticket volume grew from 2 → 28 within the first two weeks of January 2026, showing an initial surge in support requests.

2. **Average Forecast:** Predicts a stable ticket volume of 15/week, representing a conservative projection suitable for routine planning.

3. **Linear Forecast:** Projects a rapid increase in ticket volume 54 → 132 (54,80,106,132…) reflecting a growth trend in historical data and highlighting potential peak periods.

4. **Trend Divergence:** The difference between the two forecasts demonstrates the contrast between conservative planning and growth-oriented trend anticipation.

5. **Operational Insight:** Even with limited historical data, the linear model identifies possible future peaks, helping support teams anticipate workload and plan resources accordingly.

Both forecasting methods provide complementary perspectives. The **weekly average forecast** offers a simple, stable projection, while the **linear trend forecast** highlights potential spikes in ticket volume. Including both forecasts allows for **balanced operational planning**, supports proactive resource allocation, and helps ensure readiness for both typical and peak support periods.

➔ To view the python code which was applied for predicting "Ticket Volume Forecast", Click here in this link: Ticket Volume Forecast.ipynb

7. Limitations & Assumptions

- **Limited accuracy in CSAT score:** No CSAT score is provided, and so the sentiments column is used to calculate it. Sentiments are not provided by the user and are instead inferred by agents, potentially leading to incorrect sentiment estimates. The sentiments category also only reflects the customers' sentiment towards the issue, rather than satisfaction with the customer service provided, meaning we are not provided with a direct indicator of customer satisfaction.

- **Short time window:** Data was observed over the course of only one week. This makes it unsuitable for modeling long-term patterns like weekly, monthly, seasonal, etc. There is a risk of interpreting temporary fluctuations as a permanent pattern. We have to make a weak generalization about this short time period being reflective of all time trends; the analysis will fail to perform well on unseen data because it was trained on a non-representative sample.

- **Small dataset size:** We were given a limited dataset to work with. It is difficult to detect significant, true relationships because there is limited variability in the data. Due to increased sensitivity, a few outliers could significantly skew results and mislead conclusions. We may miss out on some meaningful patterns and insights due to them not being clear in our small dataset.

8. Key Observations & Insights

8.1 Challenges & Insights Identified During Dashboard Development

- The analysis covered only one week (Jan 4–10, 2026), which limited the ability to observe strong or consistent long-term trends.
- With just 30 tickets, some visuals lacked depth, and small changes had a noticeable impact on KPIs and charts.
- Several key metrics, such as resolution time and SLA compliance, required manual DAX calculations, increasing dashboard complexity.
- All tickets were treated equally, even though some issues were clearly more complex than others.
- Despite these constraints, the dashboard clearly highlighted technical and billing issues as the main drivers of ticket volume and negative sentiment.

8.2 Challenges in Insight Interpretation & Recommendations

- There was no direct CSAT score available, so customer sentiment had to be used as a proxy, which limits accuracy.
- Many tickets shared similar or overlapping tags, making it harder to clearly separate and diagnose root causes.
- High SLA compliance did not always align with customer sentiment, requiring deeper interpretation beyond surface-level metrics.
- Recommendations had to balance being realistic with being impactful, given the limited data available.
- Insights were meaningful but needed to be framed carefully to avoid over-interpreting short-term patterns.

8.3 Challenges Faced in Ticket Volume Forecasting

- The very short historical data window reduced forecast reliability and prevented identification of seasonal or long-term patterns.
- The linear forecast showed sharp growth, but results were highly dependent on assumptions and early data behavior.

- Weekly aggregation simplified the forecast but masked daily fluctuations seen in the dashboard.
- Different forecasting methods produced noticeably different results, highlighting uncertainty in projections.
- Overall, the forecasts were most useful for short-term planning rather than long-term strategic decision-making.

9. Learning Outcomes

- This week strengthened our understanding of how to design and develop interactive dashboards that effectively present customer support performance metrics and operational KPIs.
- We improved our ability to identify and interpret trends in ticket volume, categories, priorities, and sentiment, enabling deeper insight into recurring customer issues.
- Through dashboard-based analysis, we enhanced our skills in translating data-driven insights into practical and actionable recommendations aligned with business objectives.
- We developed foundational knowledge in ticket volume forecasting and gained appreciation for how predictive techniques support proactive resource allocation and workload planning.
- Overall, this week reinforced the importance of collaborative analysis, data visualization, and professional judgment in supporting informed decision-making within customer support operations.

10. Conclusion

We emphasized the importance of trend analysis in customer support analytics for transforming structured ticket data into meaningful business insights in this report. By using a cleaned and well-organized dataset along with an interactive dashboard, we successfully identified patterns related to ticket volume, issue categories, service efficiency, agent workload, and customer sentiment. These insights enabled us to uncover operational gaps, recurring customer issues, and areas that require process optimization.

Furthermore, we demonstrated how the integration of actionable recommendations and ticket volume forecasting supports proactive and informed decision-making. We highlight the value of data-driven approaches in enhancing service quality, optimizing resource allocation, and improving the overall customer experience. Overall, our analysis reinforces the role of analytics as a critical enabler of continuous improvement in customer support operations.